

# Keyush Nisar

Mumbai, India | Nisarkeyush3@gmail.com | +91 93226 43414 | linkedin.com/in/keyush-n  
github.com/nisaral

## Education

### B.Tech in Artificial Intelligence and Machine Learning

Aug 2023 – 2027

D.J. Sanghvi College of Engineering, Mumbai University

## Research Experience

### Uncertainty-Preserving Distillation for Diffusion World Models

GitHub

*Independent Research — PyTorch, Diffusion Models, MBPO, SAC*

- Investigated whether compressing diffusion-based world models into faster one-step models preserves the **epistemic uncertainty** needed for reinforcement learning, rather than only preserving next-state prediction accuracy.
- Proved that a natural single-latent teacher–student uncertainty-matching objective does not uniquely identify epistemic uncertainty; constructed counterexamples and derived a multi-latent formulation that separately constrains epistemic and aleatoric uncertainty.
- Tested the formulation through controlled and policy-driven Soft Actor-Critic experiments; after tracing and fixing a training-pipeline bug, confirmed recovery of both uncertainty components and evaluated whether improved uncertainty fidelity transfers to downstream policy performance.

### Machine Learning Engineer Research Intern — RAI Tracker

Apr 2026 – Jul 2026

*Responsible Systems Ltd, Remote, London, UK*

*Supervisor: Dr. Rajeev Chakraborty*

- Studied behavioral drift in multi-agent retrieval-augmented generation systems, using runtime telemetry to evaluate how deployed AI systems behave across defined ethical dimensions.
- Designed a three-gate validation framework covering **instrumentation validity, statistical validity, and evidentiary independence**; built OpenTelemetry instrumentation and a system-independent telemetry schema to turn runtime traces into experimentally testable behavioral signals.

### Generative AI Research Intern — CoE-CNDS Labs, VJTI

Feb 2025 – Jan 2026

*Mumbai, India*

*Supervisors: Dr. Faruk Kazi, Aum Patil*

- Fine-tuned **XLM-RoBERTa** with LoRA adapters on a 23-language Indic corpus, developing an efficient adaptation of the multilingual model for low-resource natural language processing.
- Researched unstructured video understanding by developing a semantic VideoRAG pipeline; used CLIP-based vision-language representations for keyframe extraction and designed a custom rolling-gradient similarity method for temporal matching.

## Industry Experience

### Machine Learning Intern — CerebralZip

Jul 2026 – Present

*VJTI-incubated Startup, Mumbai*

- Developing machine learning systems for a government environmental and safety monitoring platform, including multilingual incident classification, detection of multiple issues within a single report, automated routing to relevant departments, and real-time alerts from noisy citizen reports.
- Developing time-series forecasting and source-attribution models for environmental monitoring, with evaluation and inference pipelines supporting government-facing deployments and operational alert workflows.
- Prototyping **automatic license plate recognition** and computer-vision pipelines for degraded field footage, combining license-plate detection with optical character recognition under low-light, blur, and long-distance conditions.

### AI Production Engineering Intern — EnsanAI

Aug 2025 – Apr 2026

*Remote, Delaware, US*

- Designed a domain-specific healthcare AI evaluation framework covering HIPAA-oriented compliance, empathy, behavioral and data drift, and safety metrics to systematically assess model behavior across clinical workflows.
- Deployed production AI microservices on Google Cloud Platform, stress-tested inference workloads, and built a **Model Context Protocol server** connecting database and operational tools with downstream services; reduced inference costs by 35% and system latency by 40%.

## Research & Publications

### VRAG: Hierarchical Semantic Retrieval for Long-Form Video Question

2026

#### Answering

*The Visual Computer, Springer — Under Peer Review*

- Contributed to a hierarchical video retrieval system that progressively narrows a long video from **topics to scenes to keyframes**, allowing question-answering models to locate fine-grained visual evidence instead of searching the full video uniformly.
- Worked on multimodal and temporal embeddings and formulated the **Gaussian-smoothed keyframe scoring** method, which reduces isolated similarity peaks and produces more temporally stable evidence selection.
- Evaluated the system on **463 videos and 1,103 queries**, achieving **42.2% Video Recall@1** and **57.0% Scene Recall@3**, with a **3.7× improvement in Video Recall@1** over flat retrieval.

### DIO: Black-Box Online Routing for Heterogeneous vLLM Replicas

2026

*The Journal of Supercomputing, Springer — Under Review*

- Led the design of a lightweight OpenAI-compatible gateway that routes requests across inference replicas with different processing speeds, learning their performance directly from end-to-end request telemetry without access to internal model states.
- Developed an online latency model that learns the processing-time behavior of each inference replica and continuously adapts routing decisions as workloads change.
- Achieved **48.3% lower 99th-percentile latency** than Round-Robin routing under controlled 2× service-time asymmetry, with approximately **14μs** of scheduling overhead.
- Identified a limitation of learned routing: a model can correctly rank faster and slower workers while remaining poorly calibrated when its predictions are interpreted as absolute latency thresholds, motivating a separation between routing and admission decisions.

## Selected Projects

### LM-JEPA for Symbolic Regression

GitHub

*PyTorch, Deep Learning, Representation Learning*

- Researched and adapted Yann LeCun's **Joint Embedding Predictive Architecture** from computer vision to symbolic regression, learning representations of mathematical structure without autoregressive generation.
- Developed a representation-learning pipeline for extracting mathematical relationships from complex data.

### Voice-First Financial Agent

GitHub

*FastAPI, LangGraph, OpenTelemetry*

- Engineered a low-latency voice agent using WebSocket streaming and asynchronous processing, with voice-activity detection for handling user interruptions, achieving sub-500ms time-to-first-audio.
- Built OpenTelemetry-based observability to measure end-to-end latency and track state transitions across the agent's multi-stage workflow.
- Implemented asynchronous tool execution and state management to keep financial-agent interactions responsive while maintaining traceability across transitions.

## Technical Skills

**Machine Learning:** Retrieval-Augmented Generation, Transformers, Parameter-Efficient Fine-Tuning, LoRA, Diffusion Models, Uncertainty Quantification, Trustworthy Machine Learning, Large Language Model Evaluation

**Programming & Frameworks:** Python, Go, SQL, PyTorch, TensorFlow, Scikit-learn, FastAPI

**Systems & Infrastructure:** vLLM, OpenTelemetry, Kafka, gRPC, Docker, Kubernetes, CI/CD, PostgreSQL

**Cloud & Monitoring:** AWS, GCP, Arize Phoenix, Model Monitoring, Prometheus, Grafana

## Co-curricular Activities

### Technical Head, ML Mentor & Co-founder — CodeAI

Jun 2025 – Present

*ML Club at DJSCE, Mumbai*

- Mentor students in Machine Learning and Data Science and lead technical projects within the club.
- Directed a team developing an AI engine for Aspira Pathology Lab that automates prescription interpretation, achieving **92%+ accuracy** across a 3,500+ test catalog.

## Awards & Achievements

- **Nexus Kaggle Competition:** 2nd Runner-up — F1 score 99.56% in drug status prediction.
- **Hack2Infinity:** Best in AI — Mental Health App “MindCompanion”.
- **CodeShastra 12th Edition:** 2nd Runner-up — ERP-based AI FaithTech App.
- **Catnip Infotech Hackathon:** Runner-up — Recommendation System.